

1. File Transfer Calculation

Given:

File size = 150 MB

Bandwidth = 50 Mbps

Segment size = 1500 bytes

Header + Trailer = 40 bytes/frame

Calculation:

File size = $150 \times 1,000,000 = 150,000,000$ bytes

Number of segments = $150,000,000 \div 1500 = 100,000$ segments

Number of frames = 100,000 frames

Total overhead = $100,000 \times 40 = 4,000,000$ bytes (4 MB)

Total data transmitted = 150 MB + 4 MB = 154 MB

Total bits = $154 \times 8 = 1232$ Mb

Transmission time = $1232 \times 50 = 24.64$ seconds

Answer:

Segments = 100,000

Frames = 100,000

Data Link overhead = 4 MB

Transmission time = 24.64 seconds

2. Sliding Window Protocol

Given:

Window size = 6 frames

Total frames = 48

Calculation:

Number of windows = $48 \div 6 = 8$ windows

One lost frame per window = 8 retransmissions

Answer:

Transmission windows = 8

Retransmissions = 8 frames

3. IP Fragmentation

Given:

IP packet = 12,000 bytes

MTU = 1500 bytes

IP header = 20 bytes

Calculation:

Data per fragment = $1500 - 20 = 1480$ bytes

Number of fragments = $12000 \div 1480 = 9$ fragments

Header overhead = $9 \times 20 = 180$ bytes

Answer:

Fragments created = 9

Header overhead = 180 bytes

4. Sliding Window Protocol

Given:

Window size = 6 frames

Total frames = 48

Calculation:

Number of windows = $48 \div 6 = 8$ windows

One lost frame per window = 8 retransmissions

Answer:

Transmission windows = 8

Retransmissions = 8 frames

5. Session Layer Synchronization

Given:

File size = 600 MB

Checkpoint every 60 MB

Failure after 390 MB

Calculation:

Checkpoints before failure = 60, 120, 180, 240, 300, 360 = 6 checkpoints

Last checkpoint = 360 MB

Data to retransmit = 390 - 360 = 30 MB

Answer:

Checkpoints created = 6

Data retransmitted = 30 MB

6. Presentation Layer

Given:

Original file = 900 MB

Compression = 40%

Encryption overhead = 5% of compressed size

Calculation

Compressed file size = $900 \times (100 - 40)\%$

= 900×0.60

= 540 MB

Encrypted file size = $540 + (5\% \text{ of } 540)$

= $540 + 27$

= 567 MB

Total percentage reduction = $\diamond$

= 37% reduction

Answer

Compressed file size = 540 MB

Encrypted file size = 567 MB

Overall reduction = 37%

7. Application Layer

Given:

Data = 3 GB = 3000 MB

Request size = 3 MB

Throughput = 120 Mbps

Calculation

Number of requests = $3000 \div 3$

= 1000 requests

Convert data to megabits = $3000 \times 8 = 24000 \text{ Mb}$

Transmission time = $24000 \div 120$

= 200 seconds

Answer

Total requests = 1000

Transmission time = 200 seconds

8. End-to-End Delay

Given

Application = 4 ms

Transport = 3 ms

Network = 5 ms

Data Link = 2 ms

Physical = 1 ms

Propagation = 18 ms

Transmission = 12 ms

Calculation

Processing delay = $4 + 3 + 5 + 2 + 1 = 15 \text{ ms}$

Total delay = $15 + 18 + 12 = 45 \text{ ms}$

Answer

Processing delay = 15 ms

Total end-to-end delay = 45 ms

9. Protocol Overhead

Given

Useful data = 80 MB

Frame size = 1600 bytes

Overhead = 80 bytes/frame

Useful data per frame = $1600 - 80 = 1520 \text{ bytes}$

Convert data: $80 \text{ MB} = 80 \times 1024 \times 1024 = 83,886,080 \text{ bytes}$

Calculation

Number of frames = $83,886,080 \div 1520 \approx 55,189$ frames

Total overhead = $55,189 \times 80 = 4,415,120$ bytes ≈ 4.21 MB

Transmission efficiency = $(1520 \div 1600) \times 100 = 95\%$

Answer

Frames required = 55,189

Total overhead = 4,415,120 bytes (≈ 4.21 MB)

Transmission efficiency = 95%

10. Multiple OSI Layers

Given

Original data = 240 MB

Compression = 30%

Segment size = 1400 bytes

Overhead = 60 bytes/frame

Calculation

Compressed size = $240 \times 0.70 = 168$ MB

Convert to bytes 168 MB = $168 \times 1024 \times 1024 = 176,160,768$ bytes

Number of segments = $176,160,768 \div 1400 \approx 125,830$ segments

Protocol overhead = $125,830 \times 60 = 7,549,800$ bytes ≈ 7.20 MB

Final transmitted data = $176,160,768 + 7,549,800 = 183,710,568$ bytes ≈ 175.20 MB

Answer

Compressed file size = 168 MB

Segments transmitted = 125,830

Protocol overhead = 7,549,800 bytes (≈ 7.20 MB)

Final transmitted data = 183,710,568 bytes (≈ 175.20 MB)